

HNW Underwriting Automation POC

Document Classification, Extraction & Summarization

Scoping & Requirements

- Define target document types
 - 2 Application forms, English to be extracted
- Define extraction field schema per document type
 - TBD, 5-6 fields in the
- Build classification taxonomy
 - Per document for now (TBD)
- Define summarization
 - summarization of fields extracted in the document
- Establish KPIs:
 - Classification precision/recall
 - Correctness of entities
 - Completeness/Coverage

Ingestion & Preprocessing

Pipeline Setup

- Implement ingestion API or event triggers
- Store raw docs with metadata

OCR & Layout

- Evaluate different OCR engines
- Preprocess: denoise, rotate etc
 - TBD
- Extract layout: blocks, lines, tables
 - Normalization: line breaks, text cleanup etc.

Document Classification

Taxonomy

- Set primary & secondary classes
- Define unknown/other category

Baselines & Models

- Experiment with various options

Evaluation

- Confusion matrix?

Field Extraction

Schema Setup

- Define required/optional fields
- Configure validation rules

Extraction

- Rule-based (forms)
- LLM based

Post-processing

- Normalize dates, currencies, addresses
- Cross-field checks

Summarization

Requirements

- Build summary for provided documents
- Build summary of summaries - TBC if we get more documents

Extractive Summary

- Extract key facts with linked spans

Abstractive Summary

- Include citations to source text

Evaluation

Workflows

UI

- Summary with citations
- Summary of summaries
- Extracted fields
- Document viewer

Feedback

- TBD

NFR's

Config contracts

- Structured schema
- Prompt library

Data Contracts

- Define canonical JSON schemas
- Schema versoining

Storage

- DB for Structured outputs
- Audit logging

Monitoring & Observability

Drift Detection

- OCR quality?
- Classification distribution drift
- Extraction confidence shifts

Observability

- Trace IDs across components